

MODULE 1

FUNDAMENTALS OF LOGIC

Proposition

A **proposition** is a meaningful declarative sentence that clearly expresses a statement and has a definite truth value.

This means the statement must be either completely true or completely false. It cannot be both true and false at the same time.

In logic, only statements that can be judged as true or false are considered propositions. Questions, commands, and exclamations are not propositions because they do not have truth values.

Propositions are usually denoted by small letters such as p, q, r, s , etc.

If a proposition is true, we assign it the value 1. If it is false, we assign it the value 0.

Example:

p : 11 is a prime number. (True) q : 9 is an even number. (False)

Logical Connectives

Logical connectives are special words or symbols used to combine two or more propositions to form a new proposition.

They help us construct complex logical statements from simple ones.

Common logical connectives are:

- NOT
- AND
- OR
- IF... THEN
- IF AND ONLY IF

A statement formed using connectives is called a **compound proposition**. A statement without connectives is called a **simple proposition**.

Negation (NOT)

Negation reverses the truth value of a proposition.

If p is a proposition, its negation is written as $\neg p$ and read as “not p ”.

If p is true, then $\neg p$ is false. If p is false, then $\neg p$ is true.

Thus, negation always produces the opposite truth value.

p	$\neg p$
0	1
1	0

Conjunction (AND)

Conjunction joins two propositions using the word “and”.

It is written as $p \wedge q$.

The statement $p \wedge q$ is true only when both p and q are true simultaneously.

If either one of them is false, the whole statement becomes false.

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

Disjunction (OR)

Disjunction joins two propositions using the word “or”.

It is written as $p \vee q$.

The statement $p \vee q$ is true if at least one of the propositions is true.

It is false only when both p and q are false.

This is called inclusive OR because it includes the case where both are true.

p	q	$p \vee q$
0	0	0
0	1	1
1	0	1
1	1	1

Implication (IF... THEN)

Implication expresses a conditional relationship.

It is written as $p \rightarrow q$ and read as “If p , then q ”.

The implication is false only when p is true and q is false. In all other cases, it is true.

p	q	$p \rightarrow q$
0	0	1
0	1	1
1	0	0
1	1	1

Biconditional (IF AND ONLY IF)

Biconditional represents a two-way condition.

It is written as $p \leftrightarrow q$.

The statement is true when both propositions have the same truth value (both true or both false).

p	q	$p \leftrightarrow q$
0	0	1
0	1	0
1	0	0
1	1	1

Exclusive OR

Exclusive OR means exactly one of the propositions is true.

It is written as $p \nabla q$.

If both are true or both are false, it becomes false.

p	q	$p \nabla q$
0	0	0
0	1	1
1	0	1
1	1	0

Complete Combined Truth Table

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \nabla q$
0	0	1	0	0	1	1	0
0	1	1	0	1	1	0	1
1	0	0	0	1	0	0	1
1	1	0	1	1	1	1	0

Important Logical Terms

Tautology:

A compound proposition that is true for every possible combination of truth values of its components is called a tautology.

In other words, no matter what truth values we assign to the variables, the statement always remains true.

Example:

The statement $p \vee \neg p$ is a tautology.

p	$\neg p$	$p \vee \neg p$
0	1	1
1	0	1

Since the final column is always 1, it is a tautology.

Contradiction:

A compound proposition that is false for every possible combination of truth values is called a contradiction.

That means the statement never becomes true under any situation.

Example:

The statement $p \wedge \neg p$ is a contradiction.

p	$\neg p$	$p \wedge \neg p$
0	1	0
1	0	0

Since the final column is always 0, it is a contradiction.

Contingency:

A compound proposition that is true in some cases and false in some cases is called a contingency. It is neither always true nor always false.

Example:

The statement $p \wedge q$ is a contingency.

p	q	$p \wedge q$
0	0	0
0	1	0
1	0	0
1	1	1

Since the final column contains both 0 and 1, it is a contingency.

Logical Equivalence:

Two propositions are said to be logically equivalent if they have exactly the same truth values for all possible combinations of truth values of their variables.

If $p \leftrightarrow q$ is a tautology, then p and q are logically equivalent.

Example:

The statements $p \rightarrow q$ and $\neg p \vee q$ are logically equivalent.

p	q	$p \rightarrow q$	$\neg p \vee q$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

Since the last two columns are identical, the two propositions are logically equivalent.

Problems on Tautology**Problem 1**

P.T. for any proposition p, q, r the compound proposition

$$[(p \vee q) \rightarrow r] \leftrightarrow [\sim r \rightarrow \sim (p \vee q)]$$

is a tautology.

Sol:

p	q	r	$p \vee q$	$(p \vee q) \rightarrow r$	$\sim r$	$\sim (p \vee q)$	$\sim r \rightarrow \sim (p \vee q)$	$x \leftrightarrow y$
1	1	1	1	1	0	0	1	1
1	1	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	1
1	0	0	1	0	1	0	0	1
0	1	1	1	1	0	0	1	1
0	1	0	1	0	1	0	0	1
0	0	1	0	1	0	1	1	1
0	0	0	0	1	1	1	1	1

Since the final column $x \leftrightarrow y$ contains only 1's,

$$[(p \vee q) \rightarrow r] \leftrightarrow [\sim r \rightarrow \sim (p \vee q)]$$

is a tautology.

Problem 2

P.T. for any proposition p, q, r the compound proposition

$$[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$$

is a tautology.

Sol:

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)$	$s \rightarrow r$
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	1
1	0	1	1	1	1	1	1
1	0	0	1	0	1	0	1
0	1	1	1	1	1	1	1
0	1	0	1	1	0	0	1
0	0	1	0	1	1	0	1
0	0	0	0	1	1	0	1

Since the final column contains only 1's, Hence tautology.

Problem 3

P.T. for any proposition p, q, r the compound proposition

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow q)$$

is a tautology.

Sol:

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$x \wedge y \rightarrow x$
0	0	0	1	1	1	1
0	0	1	1	1	1	1
0	1	0	1	0	0	1
0	1	1	1	1	1	1
1	0	0	0	1	0	1
1	0	1	0	1	0	1
1	1	0	1	0	0	1
1	1	1	1	1	1	1

Since the final column contains only 1's, Hence tautology.

Problem 4

P.T. for any proposition p, q, r the compound proposition

$$[p \rightarrow (q \rightarrow r)] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$$

is a tautology.

Sol:

p	q	r	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$	$p \rightarrow q$	$p \rightarrow r$	$(p \rightarrow q) \rightarrow (p \rightarrow r)$	$x \rightarrow y$
0	0	0	1	1	1	1	1	1
0	0	1	1	1	1	1	1	1
0	1	0	0	1	1	1	1	1
0	1	1	1	1	1	1	1	1
1	0	0	1	1	0	0	1	1
1	0	1	1	1	0	1	1	1
1	1	0	0	0	1	0	0	1
1	1	1	1	1	1	1	1	1

Since the final column contains only 1's, Hence tautology.

Problem 5

P.T. for any proposition p, q, r the compound proposition

$$(p \vee q) \wedge [(p \rightarrow q) \wedge (q \rightarrow r)]$$

Sol:

p	q	r	$p \vee q$	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$x \wedge y$
0	0	0	0	1	1	1	0
0	0	1	0	1	1	1	0
0	1	0	1	1	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	0	1	0	0
1	0	1	1	0	1	0	0
1	1	0	1	1	0	0	0
1	1	1	1	1	1	1	1

Since the final column does not contain all 1's, is not a tautology.

Logical Equivalence**Problem 1**

Verify by truth table that

$$\sim (p \vee q \vee r) \equiv \sim p \wedge \sim q \wedge \sim r$$

Sol:

p	q	r	$p \vee q \vee r$	$\sim (p \vee q \vee r)$	$\sim p \wedge \sim q$	$\sim p \wedge \sim q \wedge \sim r$
0	0	0	0	1	1	1
0	0	1	1	0	1	0
0	1	0	1	0	0	0
0	1	1	1	0	0	0
1	0	0	1	0	0	0
1	0	1	1	0	0	0
1	1	0	1	0	0	0
1	1	1	1	0	0	0

Since the columns for $\sim (p \vee q \vee r)$ and $\sim p \wedge \sim q \wedge \sim r$ are identical, Hence proved.

Problem 2

Verify by truth table that

$$[(p \vee q) \rightarrow r] \equiv [\sim r \rightarrow \sim (p \vee q)]$$

Sol:

p	q	r	$p \vee q$	$(p \vee q) \rightarrow r$	$\sim r$	$\sim r \rightarrow \sim (p \vee q)$
0	0	0	0	1	1	1
0	0	1	0	1	0	1
0	1	0	1	0	1	0
0	1	1	1	1	0	1
1	0	0	1	0	1	0
1	0	1	1	1	0	1
1	1	0	1	0	1	0
1	1	1	1	1	0	1

Since the columns for $(p \vee q) \rightarrow r$ and $\sim r \rightarrow \sim (p \vee q)$ are identical, Hence proved.

Laws of Logic

In propositional logic, several fundamental laws help us simplify logical expressions and prove equivalence between propositions. These laws are universally valid and are used frequently in mathematical proofs and logical reasoning.

Let T_0 denote a **tautology** (always true statement) and let F_0 denote a **contradiction** (always false statement).

1. Law of Double Negation

For any proposition p ,

$$\sim (\sim p) \equiv p$$

Explanation:

Negation reverses the truth value of a proposition. If we negate a statement twice, the original truth value is restored. Thus, double negation cancels itself and gives back the original proposition.

2. Idempotent Laws

For any proposition p ,

$$(p \vee p) \equiv p$$

$$(p \wedge p) \equiv p$$

Explanation:

Repeating the same proposition using OR or AND does not change its value. A statement combined with itself remains unchanged.

3. Identity Laws

For any proposition p ,

$$(p \vee F_0) \equiv p$$

$$(p \wedge T_0) \equiv p$$

Explanation:

In OR operation, false behaves like 0 in addition. In AND operation, true behaves like 1 in multiplication. Hence, combining with these identity elements does not change the proposition.

4. Inverse Laws

For any proposition p , $(p \vee \sim p) \equiv T_0$

$$(p \wedge \sim p) \equiv F_0$$

Explanation:

A statement OR its negation is always true. A statement AND its negation is always false.

5. Domination Laws

For any proposition p ,

$$(p \vee T_0) \equiv T_0$$

$$(p \wedge F_0) \equiv F_0$$

Explanation:

If a statement is ORed with a true statement, the result is always true. If a statement is ANDed with a false statement, the result is always false.

6. Commutative Laws

For any propositions p, q ,

$$(p \vee q) \equiv (q \vee p)$$

$$(p \wedge q) \equiv (q \wedge p)$$

Explanation:

The order of propositions does not affect the result in OR and AND operations.

7. Absorption Laws

For any propositions p, q ,

$$[p \vee (p \wedge q)] \equiv p$$

$$[p \wedge (p \vee q)] \equiv p$$

Explanation:

When a proposition appears both inside and outside a compound expression, the larger expression simplifies to the proposition itself.

8. De Morgan's Laws

For any propositions p, q ,

$$\sim (p \vee q) \equiv \sim p \wedge \sim q$$

$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

Explanation:

Negation distributes over OR and AND by changing the operator and negating each component.

9. Associative Laws

For any propositions p, q, r ,

$$p \vee (q \vee r) \equiv (p \vee q) \vee r$$

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$$

Explanation:

Grouping of propositions does not affect the result in OR and AND operations.

10. Distributive Laws

For any propositions p, q, r ,

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

Explanation:

These laws allow distribution of one logical operator over another, similar to algebraic distributive property.

Important Logical Equivalences

For any propositions p, q ,

$$(p \rightarrow q) \equiv \sim p \vee q$$

$$\sim (p \rightarrow q) \equiv (p \wedge \sim q)$$

Explanation:

An implication can be rewritten using OR and negation. Negation of implication produces conjunction.

Transitive Rule

The transitive rule allows logical equivalence to be transferred through an intermediate proposition.

If

$$u \equiv v \quad \text{and} \quad v \equiv w,$$

then

$$u \equiv w.$$

Explanation:

If one statement is equivalent to a second statement, and the second statement is equivalent to a third statement, then the first and third statements must also be equivalent.

This rule is useful for step-by-step simplification.

Substitution Rule

If $p \equiv q$, and p appears in a compound proposition u , then replacing p with q produces a new proposition v such that

$$u \equiv v.$$

Explanation:

Equivalent expressions can be substituted without changing the logical value of the statement.

Dual of a Statement

Definition

Let s be a statement. If s contains no logical connectives other than $\wedge$ and $\vee$, then the **dual** of s , denoted by s^d , is the statement obtained from s by:

- Replacing each occurrence of $\wedge$ by $\vee$
- Replacing each occurrence of $\vee$ by $\wedge$
- Replacing each occurrence of T_0 (True) by F_0 (False)
- Replacing each occurrence of F_0 (False) by T_0 (True)

Example

If

$$u : p \wedge (q \vee \neg r) \vee (s \wedge T_0),$$

then its dual is

$$u^d : p \vee (q \wedge \neg r) \wedge (s \vee F_0).$$

Remarks

1. $(p^d)^d \iff p$
(i.e., the dual of the dual of p is logically equivalent to p)

2. If $p \iff q$, then $p^d \iff q^d$
(Principle of Duality)

- 3.

$$(\neg p)^d \iff \neg p$$

$$(p \vee \neg p)^d \iff (p \wedge \neg p)$$

$$(p \wedge \neg p)^d \iff (p \vee \neg p)$$

Negation of a Proposition

Problem 1

Let x be a specified number. Write down the negation of the following proposition:

”If x is not a real number, then it is not a rational number and not an irrational number.”

Solution

Let us define the following propositions:

p : x is a real number

q : x is a rational number

r : x is an irrational number

The given statement can be written symbolically as:

$$\neg p \rightarrow (\neg q \wedge \neg r)$$

Negation of the Proposition

To find the negation of the given implication:

$$\neg [\neg p \rightarrow (\neg q \wedge \neg r)]$$

Using the logical identity:

$$\neg(A \rightarrow B) \equiv A \wedge \neg B$$

We get:

$$\neg p \wedge \neg(\neg q \wedge \neg r)$$

Using De Morgan's Law:

$$\neg(\neg q \wedge \neg r) \equiv \neg\neg q \vee \neg\neg r$$

$$\equiv q \vee r$$

Therefore, the negation becomes:

$$\neg p \wedge (q \vee r)$$

The negation of the given proposition is:

” x is not a real number and it is either a rational number or an irrational number.”

Problem 2

Express each of the following statements in symbolic form. Then find their negations and write them in smooth English.

1. Vimala will get a good education if she puts her studies before her interest in cheer leading.
2. Nirma is doing her homework and Kamala is practicing her music lesson.

Solution

Let us define the following propositions:

p : Vimala puts her studies before her interest in cheer leading

q : Vimala will get a good education

r : Nirma is doing her homework

s : Kamala is practicing her music lesson

Symbolic Form

1. $p \rightarrow q$

2. $r \wedge s$

Negations

1.

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

Negation in English:

Vimala puts her studies before her interest in cheer leading and she will not get a good education.

2.

$$\neg(r \wedge s) \equiv \neg r \vee \neg s$$

Negation in English:

Nirma is not doing her homework or Kamala is not practicing her music lesson.

Prove Without Using Truth Tables

Show that

$$\{(p \rightarrow q) \rightarrow r\} \leftrightarrow \{(\neg p \vee q) \rightarrow r\}$$

is a tautology.

Solution

Note: To prove $U \leftrightarrow V$ is a tautology, it is enough to prove $U \equiv V$ (by logical equivalence).

We know that:

$$p \rightarrow q \equiv \neg p \vee q$$

Therefore,

$$(p \rightarrow q) \rightarrow r \equiv (\neg p \vee q) \rightarrow r$$

Hence,

$$\{(p \rightarrow q) \rightarrow r\} \leftrightarrow \{(\neg p \vee q) \rightarrow r\}$$

is a tautology.

Simplify the Following Using Laws of Logic

(i) $(p \vee q) \wedge [\neg\{\neg p \wedge q\}]$

Solution:

$$(p \vee q) \wedge \neg(\neg p \wedge q)$$

Using De Morgan's Law:

$$= (p \vee q) \wedge (\neg\neg p \vee \neg q)$$

Using Double Negation:

$$= (p \vee q) \wedge (p \vee \neg q)$$

Using Distributive Law:

$$= p \vee (q \wedge \neg q)$$

Using Inverse Law:

$$= p \vee F_0$$

Using Identity Law:

$$= p$$

(ii) $\neg[\neg\{(p \vee q) \wedge r\} \vee \neg q]$

Solution:

$$\neg[\neg\{(p \vee q) \wedge r\} \vee \neg q]$$

Using De Morgan's Law:

$$= \neg[\neg\{(p \vee q) \wedge r\}] \wedge \neg(\neg q)$$

Using Double Negation:

$$= ((p \vee q) \wedge r) \wedge q$$

Using Associative Law:

$$= (p \vee q) \wedge (r \wedge q)$$

Using Commutative Law:

$$= (p \vee q) \wedge (q \wedge r)$$

Using Associative Law:

$$= [(p \vee q) \wedge q] \wedge r$$

Using Absorption Law:

$$= q \wedge r$$

Final Answers:

(i) p

(ii) $q \wedge r$

Proof of Logical Equivalences Without Truth Tables

(I) $p \rightarrow (q \rightarrow r) \iff (p \wedge q) \rightarrow r$

$$p \rightarrow (q \rightarrow r) = \neg p \vee (q \rightarrow r) = \neg p \vee (\neg q \vee r) = \neg p \vee \neg q \vee r$$

$$(p \wedge q) \rightarrow r = \neg(p \wedge q) \vee r = (\neg p \vee \neg q) \vee r = \neg p \vee \neg q \vee r$$

$$\therefore p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

(II) $[p \vee q \vee (\neg p \wedge \neg q \wedge r)] \iff (p \vee q \vee r)$

$$p \vee q \vee (\neg p \wedge \neg q \wedge r) = (p \vee q) \vee [(\neg p \wedge \neg q) \wedge r]$$

$$= (p \vee q) \vee [\neg(p \vee q) \wedge r]$$

Using distributive law:

$$A \vee (\neg A \wedge r) = (A \vee \neg A) \wedge (A \vee r)$$

Let $A = (p \vee q)$,

$$= [(p \vee q) \vee \neg(p \vee q)] \wedge [(p \vee q) \vee r]$$

$$= 1 \wedge (p \vee q \vee r) = p \vee q \vee r$$

$$\therefore [p \vee q \vee (\neg p \wedge \neg q \wedge r)] \equiv (p \vee q \vee r)$$

(III) $[(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \iff p \wedge q$

$$(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r) = \neg(\neg p \vee \neg q) \vee (p \wedge q \wedge r)$$

$$= (p \wedge q) \vee (p \wedge q \wedge r)$$

Using absorption law:

$$= p \wedge q$$

$$\therefore [(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \equiv p \wedge q$$

$$(IV) (p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \iff \neg(q \vee p)$$

$$(p \rightarrow q) = \neg p \vee q$$

So LHS:

$$(\neg p \vee q) \wedge [\neg q \wedge (r \vee \neg q)]$$

$$= (\neg p \vee q) \wedge \neg q$$

Since $\neg q \wedge (r \vee \neg q) = \neg q$ (absorption)

$$= (\neg p \wedge \neg q) \vee (q \wedge \neg q)$$

$$= \neg p \wedge \neg q$$

Using De Morgan's Law:

$$= \neg(p \vee q)$$

$$\therefore (p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \equiv \neg(p \vee q)$$

$$(V) [\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)] \iff r$$

$$[\neg p \wedge \neg q \wedge r] \vee [(q \wedge r) \vee (p \wedge r)]$$

Factor r :

$$= r \wedge [(\neg p \wedge \neg q) \vee q \vee p]$$

$$= r \wedge [(\neg p \wedge \neg q) \vee (p \vee q)]$$

Since $(\neg p \wedge \neg q) = \neg(p \vee q)$,

$$= r \wedge [\neg(p \vee q) \vee (p \vee q)]$$

$$= r \wedge 1 = r$$

$$\therefore [\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)] \equiv r$$

Duality

Examples

Write down the duals of the following propositions:

1. $\neg(p \vee q) \wedge [p \vee \neg(q \wedge \neg s)]$
2. $(p \wedge q) \vee [(\neg p \vee q) \wedge (\neg r \vee s)] \vee (r \wedge s)$
3. $(p \wedge \neg q) \vee (r \wedge T_0)$
4. $[(p \vee T_0) \wedge (q \vee F_0)] \vee [(r \wedge s) \wedge T_0]$

Duals

By using the definition of the dual, we find that the duals are:

1. $\neg(p \wedge q) \vee [p \wedge \neg(q \vee \neg s)]$
2. $(p \vee q) \wedge [(\neg p \wedge q) \vee (\neg r \wedge s)] \wedge (r \vee s)$
3. $(p \vee \neg q) \wedge (r \vee F_0)$
4. $[(p \wedge F_0) \vee (q \wedge T_0)] \wedge [(r \vee s) \vee F_0]$

Prove that $[(\neg p \vee q) \wedge (p \wedge (p \wedge q))] \Leftrightarrow p \wedge q$ **Hence deduce that**
 $[(\neg p \wedge q) \vee (p \vee (p \vee q))] \Leftrightarrow p \vee q$

Proof

We have

$$\begin{aligned}
 & (\neg p \vee q) \wedge (p \wedge (p \wedge q)) \\
 \Leftrightarrow & (\neg p \vee q) \wedge ((p \wedge p) \wedge q) \\
 \Leftrightarrow & (\neg p \vee q) \wedge (p \wedge q) \\
 \Leftrightarrow & [(\neg p \wedge (p \wedge q)) \vee (q \wedge (p \wedge q))] \\
 \Leftrightarrow & (F_0 \wedge q) \vee (q \wedge p) \\
 \Leftrightarrow & F_0 \vee (p \wedge q) \\
 \Leftrightarrow & p \wedge q
 \end{aligned}$$

Verify the principle of duality for the following logical equivalence:

$$[\neg(p \wedge q) \rightarrow \neg p \vee (\neg p \vee q)] \Leftrightarrow (\neg p \vee q)$$

The given logical equivalence is $u \Leftrightarrow v$, where

$$u = \neg(p \wedge q) \rightarrow (\neg p \vee (\neg p \vee q)) \quad \text{and} \quad v = \neg p \vee q$$

We note that

$$u \Leftrightarrow \neg\neg(p \wedge q) \vee (\neg p \vee (\neg p \vee q))$$

$$\Leftrightarrow (p \wedge q) \vee (\neg p \vee (\neg p \vee q))$$

Therefore,

$$u^d \Leftrightarrow (p \vee q) \wedge (\neg p \wedge (\neg p \wedge q))$$

$$\Leftrightarrow (p \vee q) \wedge (\neg p \wedge q)$$

$$\Leftrightarrow [p \wedge (\neg p \wedge q)] \vee [q \wedge (\neg p \wedge q)]$$

$$\Leftrightarrow (F_0 \wedge q) \vee (q \wedge \neg p)$$

$$\Leftrightarrow F_0 \vee (q \wedge \neg p)$$

$$\Leftrightarrow q \wedge \neg p$$

Also,

$$v^d \Leftrightarrow \neg p \wedge q \Leftrightarrow q \wedge \neg p$$

We observe that

$$u^d \Leftrightarrow v^d$$

This verifies the principle of duality for the given logical equivalence.

Logical Implication

The conditional statement $p \rightarrow q$ where p and q are propositions is called a **logical implication**. It means “If p then q ”. Such statements are also called **hypothetical** or **implicative statements**.

Note

1. When a hypothetical statement $p \rightarrow q$ is such that q is true whenever p is true, then we say that p logically implies q .

Symbolically

$$p \Rightarrow q$$

2. When a hypothetical statement $p \rightarrow q$ is such that q is not necessarily true whenever p is true, then we say that p does not imply q .

Symbolically

$$p \nRightarrow q$$

Example 1

Prove that $[(p \rightarrow q) \wedge \neg q] \Rightarrow \neg p$

Truth Table

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$(p \rightarrow q) \wedge \neg q$
0	0	1	1	1	1
0	1	1	0	1	0
1	0	0	1	0	0
1	1	0	0	1	0

From the table, when $(p \rightarrow q) \wedge \neg q$ is true, $\neg p$ is also true.

$$\bullet [(p \rightarrow q) \wedge \neg q] \Rightarrow \neg p$$

Rules of Inference

Consider a set of propositions $p_1, p_2, \dots, p_n$ and a proposition Q . Then a compound proposition of the form

$$(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \rightarrow Q$$

is called an **argument**. Here $p_1, p_2, \dots, p_n$ are called the **premises** of the argument and Q is called the **conclusion**.

Tabular Form of an Argument

It is a common practice to write the argument in the following tabular form:

$$\begin{array}{c} p_1 \\ p_2 \\ p_3 \\ \vdots \\ p_n \\ \hline \therefore Q \end{array}$$

Here the symbol $\therefore$ (three dots) stands for “**Therefore**”.

Validity of an Argument

The preceding argument is said to be **valid** if whenever each of the premises $p_1, p_2, \dots, p_n$ is true, then the conclusion Q is also true.

In other words, the argument

$$(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \rightarrow Q$$

is valid when

$$(p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n) \Rightarrow Q$$

Note:

In an argument, the premises are always taken to be **true**, whereas the conclusion may be true or false. The conclusion becomes true only when the argument is **valid**.

There exist rules of logic which can be used to establish the validity of arguments. These rules are called the **Rules of Inference**.

Rules of Inference

There exist rules of logic which can be employed for establishing the validity of arguments. These rules are called the **Rules of Inference**. Some important rules are given below.

1. Rule of Conjunctive Simplification

This rule states that, for any two propositions p and q , if $p \wedge q$ is true then p is true; i.e.,

$$(p \wedge q) \Rightarrow p$$

This rule follows from the definition of conjunction.

2. Rule of Disjunctive Amplification

This rule states that, for any two propositions p and q , if p is true then $p \vee q$ is true; i.e.,

$$p \Rightarrow (p \vee q)$$

This rule follows from the definition of disjunction.

3. Rule of Syllogism

This rule states that for any three propositions p , q , and r , if $p \rightarrow q$ is true and $q \rightarrow r$ is true, then $p \rightarrow r$ is true; i.e.,

$$\{(p \rightarrow q) \wedge (q \rightarrow r)\} \Rightarrow (p \rightarrow r)$$

This rule follows from the tautology

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

In tabular form the rule is expressed as

$$\begin{array}{c} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

4. Modus Ponens (Rule of Detachment)

This rule states that if p is true and $p \rightarrow q$ is true, then q is true; i.e.,

$$\{p \wedge (p \rightarrow q)\} \Rightarrow q$$

In tabular form,

$$\begin{array}{c} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

5. Modus Tollens

This rule states that if $p \rightarrow q$ is true and q is false, then p is false; i.e.,

$$\{(p \rightarrow q) \wedge \neg q\} \Rightarrow \neg p$$

In tabular form,

$$\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$$

6. Rule of Disjunctive Syllogism

This rule states that if $p \vee q$ is true and p is false, then q is true; i.e.,

$$\{(p \vee q) \wedge \neg p\} \Rightarrow q$$

In tabular form,

$$\frac{p \vee q \quad \neg p}{\therefore q}$$

7. Rule of Contradiction

This rule states that if $\neg p \rightarrow F_0$ is true, then p is true; i.e.,

$$(\neg p \rightarrow F_0) \Rightarrow p$$

Here F_0 denotes a contradiction (a proposition which is always false).
The rule can also be expressed as

$$\frac{\neg p \rightarrow F_0}{\therefore p}$$

Questions

1. Test the validity of the argument

Statement:

If Sachin hits a century, then he gets a car.

Sachin hits a century.

$\therefore$ Sachin gets a car.

Solution:

Let:

p : Sachin hits a century, q : Sachin gets a car

$$\begin{array}{r} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Rule used: **Modus Ponens**

Conclusion: Valid

2. Test the validity of the argument

Statement:

If Sachin hits a century, then he gets a car.

Sachin does not get a car.

$\therefore$ Sachin has not hit a century.

Solution:

$$\begin{array}{r} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

Rule used: **Modus Tollens**

Conclusion: Valid

3. Test the validity using truth table

Statement:

If Sachin hits a century, then he gets a car.

Sachin gets a car.

$\therefore$ Sachin hits a century.

Solution:

$$\begin{array}{r} p \rightarrow q \\ q \\ \hline \therefore p \end{array}$$

This is not a valid rule of inference.

Truth Table:

p	q	$p \rightarrow q$	$((p \rightarrow q) \wedge q) \rightarrow p$
0	0	1	1
0	1	1	0
1	0	0	1
1	1	1	1

Since the final column is not always 1, it is not a tautology.

Conclusion: Invalid

4. Test the validity of the argument

Statement:

If Ravi studies, then he will pass DMS paper.

If Ravi does not play cricket, then he will study.

Ravi failed in DMS paper.

$\therefore$ Ravi played cricket.

Solution:

Let:

p : Ravi studies, q : Ravi passes, r : Ravi plays cricket

$$\begin{array}{l}
 p \rightarrow q \\
 \neg r \rightarrow p \\
 \neg q \\
 \hline
 \therefore r
 \end{array}$$

Justification:

- From $p \rightarrow q$ and $\neg r \rightarrow p \Rightarrow \neg r \rightarrow q$ (Syllogism)
- From $\neg r \rightarrow q$ and $\neg q \Rightarrow r$ (Modus Tollens + Double Negation)

Conclusion: Valid

5. Test the validity of the argument Given:

If a triangle has two equal sides, then it is isosceles

If a triangle is isosceles, then it has two equal angles

Triangle ABC does not have two equal angles

$\therefore$ Triangle ABC does not have two equal sides

Let:

p : Triangle has two equal sides

q : Triangle is isosceles

r : Triangle has two equal angles

Symbolic Form:

$$1. \quad p \rightarrow q$$

$$2. \quad q \rightarrow r$$

$$3. \quad \neg r$$

To Prove: $\neg p$

Proof: Proof:

$$p \rightarrow r \quad (\text{Hypothetical Syllogism})$$

$$\neg p \quad (\text{by Modus Tollens})$$

Conclusion:

$$\therefore \neg p$$

6. Test the validity of the argument

If Ram studies, then he will pass 12th.

If Ram passes 12th, then his father gives him a bike.

If Ram does not play video games, then he will pass 12th.

Ram did not get a bike.

$$\therefore \text{Ram played video games}$$

Let:

p : Ram studies

q : Ram passes 12th

r : Ram gets a bike

v : Ram plays video games

Premises:

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\neg v \rightarrow q$$

$$\neg r$$

Conclusion:

$$\therefore v$$

7. Test the validity of the argument

$$\begin{array}{l}
 1. p \rightarrow r \\
 2. q \rightarrow r \\
 \hline
 (p \vee q) \rightarrow r
 \end{array}$$

Solution

Convert implications (Law of Implication)

$$p \rightarrow r \equiv \neg p \vee r \quad q \rightarrow r \equiv \neg q \vee r$$

Combine both statements

$$(\neg p \vee r) \wedge (\neg q \vee r)$$

Apply Distributive Law

$$(\neg p \wedge \neg q) \vee r$$

Apply De Morgan's Law

$$\neg(p \vee q) \vee r$$

Convert back to implication

$$\neg(p \vee q) \vee r \equiv (p \vee q) \rightarrow r$$

8. Test the validity of the argument

$$\begin{array}{l}
 p \rightarrow q \\
 r \rightarrow s \\
 p \vee r \\
 \hline
 \therefore q \vee s
 \end{array}$$

Solution

$$(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)$$

$$\equiv (p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg p \rightarrow r)$$

$$\equiv (p \rightarrow q) \wedge (\neg p \rightarrow r) \wedge (r \rightarrow s) \quad (\text{Commutative Law})$$

$$\Rightarrow (p \rightarrow q) \wedge (\neg p \rightarrow s) \quad (\text{Rule of Syllogism})$$

$$\Leftrightarrow (\neg q \rightarrow \neg p) \wedge (\neg p \rightarrow s) \quad (\text{Contrapositive})$$

$$\Rightarrow (\neg q \rightarrow s) \quad (\text{Rule of Syllogism})$$

$$\equiv \neg(\neg q) \vee s \quad (\text{Implication Law})$$

$$\equiv q \vee s \quad (\text{Double Negation})$$

Conclusion:

$$\therefore q \vee s$$

9. Test the validity of the argument

$$\begin{array}{l}
 p \rightarrow q \\
 r \rightarrow s \\
 \hline
 \neg q \vee \neg s \\
 \hline
 \therefore \neg(p \wedge r)
 \end{array}$$

Solution:

$$\begin{aligned}
 & (p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s) \\
 \equiv & (p \rightarrow q) \wedge (r \rightarrow s) \wedge (q \rightarrow \neg s) \quad (\text{Implication Law}) \\
 \equiv & (p \rightarrow q) \wedge (q \rightarrow \neg s) \wedge (r \rightarrow s) \quad (\text{Commutativity}) \\
 \Rightarrow & (p \rightarrow \neg s) \wedge (r \rightarrow s) \quad (\text{Rule of Syllogism}) \\
 \Leftrightarrow & (p \rightarrow \neg s) \wedge (\neg s \rightarrow \neg r) \quad (\text{Contrapositive}) \\
 \Rightarrow & (p \rightarrow \neg r) \quad (\text{Rule of Syllogism}) \\
 \equiv & \neg p \vee \neg r \quad (\text{Implication Law}) \\
 \equiv & \neg(p \wedge r) \quad (\text{De Morgan's Law})
 \end{aligned}$$

Conclusion:

$$\therefore \neg(p \wedge r)$$

10. Test the validity of the argument

$$\begin{array}{l}
 p \rightarrow q \\
 q \rightarrow r \\
 \hline
 \neg r \\
 \hline
 \therefore \neg p
 \end{array}$$

Solution:

$$\begin{aligned}
 & (p \rightarrow q) \wedge (q \rightarrow r) \wedge \neg r \\
 \Rightarrow & (p \rightarrow r) \wedge \neg r \quad (\text{Rule of Syllogism}) \\
 \Rightarrow & \neg p \quad (\text{Modus Tollens})
 \end{aligned}$$

Conclusion:

$$\therefore \neg p$$

11. Test the validity of the argument

$$\begin{array}{l}
 p \\
 p \rightarrow q \\
 s \vee r \\
 \frac{r \rightarrow \neg q}{\therefore s \vee t}
 \end{array}$$

Solution:

$$\begin{aligned}
 & p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \neg q) \\
 & \equiv p \wedge (s \vee r) \wedge (q \rightarrow r) \wedge (r \rightarrow \neg q) \\
 & \Rightarrow p \wedge (s \vee r) \wedge (q \rightarrow \neg q) \quad (\text{Syllogism}) \\
 & \Rightarrow p \wedge (s \vee r) \wedge \neg q \\
 & \Rightarrow s \quad (\text{Modus Ponens}) \\
 & \Rightarrow s \vee t \quad (\text{Addition})
 \end{aligned}$$

Conclusion:

$$\therefore s \vee t$$

12. Test the validity of the argument

$$\begin{array}{l}
 (p \vee q) \rightarrow r \\
 r \rightarrow (s \vee t) \\
 \frac{\neg(s \vee t)}{\therefore \neg(p \vee q)}
 \end{array}$$

Solution:

$$\begin{aligned}
 & [(p \vee q) \rightarrow r] \wedge [r \rightarrow (s \vee t)] \wedge \neg(s \vee t) \\
 & \Rightarrow [(p \vee q) \rightarrow (s \vee t)] \wedge \neg(s \vee t) \quad (\text{Rule of Syllogism}) \\
 & \Rightarrow \neg(p \vee q) \quad (\text{Modus Tollens})
 \end{aligned}$$

Conclusion:

$$\therefore \neg(p \vee q)$$

Open Statements and Quantifiers

Open Statement

An **open statement** is a mathematical statement that contains one or more variables.

Example:

$$x > 2, \quad x^2 = 9, \quad x + 5 = 8$$

These are not propositions unless the variable is specified.

Notation: Open statements are denoted by:

$$p(x), q(x), r(x)$$

Quantifiers

Words like *all*, *every*, *some*, *there exists* are called **quantifiers**.

Types of Quantifiers:

1. Universal Quantifier ($\forall$)

$$\forall x \in S, p(x)$$

Means: $P(x)$ is true for every element in the set.

Example:

$$\forall x \in \mathbb{R}, x + 0 = x$$

2. Existential Quantifier ($\exists$)

$$\exists x \in S, p(x)$$

Means: $P(x)$ is true for at least one element.

Example:

$$\exists x \in \mathbb{N}, x^2 = 9$$

Rules

1. Universal Specification:

$$\forall x \in S, p(x) \Rightarrow p(a)$$

2. Universal Generalization:

$$p(a) \Rightarrow \forall x, p(x)$$

Negation Rules

$$\neg(\forall x p(x)) \equiv \exists x \neg p(x)$$

$$\neg(\exists x p(x)) \equiv \forall x \neg p(x)$$

Quantifiers with Binary Propositions

$$(\forall x) p(x, y) \equiv \forall x \forall y p(x, y)$$

$$(\exists x) p(x, y) \equiv \exists x \exists y p(x, y)$$

Logical Equivalence

Two statements are said to be **logically equivalent** if they have the same truth values in all possible cases.

Basic Logical Equivalences

1. Conjunction with Universal Quantifier

$$\forall x, [p(x) \wedge q(x)] \equiv (\forall x, p(x)) \wedge (\forall x, q(x))$$

2. Disjunction with Existential Quantifier

$$\exists x, [p(x) \vee q(x)] \equiv (\exists x, p(x)) \vee (\exists x, q(x))$$

3. Implication with Existential Quantifier

$$\exists x, [p(x) \rightarrow q(x)] \equiv \exists x, [\neg p(x) \vee q(x)]$$

Rule for Negation of a quantified Statement

Statement	Negation
$\forall x P(x)$	$\exists x \neg P(x)$
$\forall x \neg P(x)$	$\exists x P(x)$
$\exists x P(x)$	$\forall x \neg P(x)$
$\exists x \neg P(x)$	$\forall x P(x)$

Important Result

Statements of the form:

$$\forall x \neg p(x)$$

are logically equivalent to:

“For no x , $p(x)$ ”

$$\forall x \neg p(x) \equiv \neg \exists x p(x)$$

Rules Involving Quantifiers and Inference

These rules are used to derive logical conclusions from given statements. They are very important in mathematical reasoning and proofs.

1. Rule of Conjunctive Simplification

If two statements are connected by AND, we can take any one of them.

$$\frac{p(x) \wedge q(x)}{\therefore p(x)}$$

2. Rule of Disjunctive Simplification

If at least one of the statements is true (OR), we can consider one part.

$$\frac{p(x) \vee q(x)}{\therefore p(x)}$$

3. Rule of Syllogism

If one statement implies another, and the second implies a third, then the first implies the third.

$$\frac{\begin{array}{l} p(x) \rightarrow q(x) \\ q(x) \rightarrow r(x) \end{array}}{\therefore p(x) \rightarrow r(x)}$$

4. Modus Ponens

If a statement is true and it implies another, then the second statement is also true.

$$\frac{\begin{array}{l} p(x) \rightarrow q(x) \\ p(x) \end{array}}{\therefore q(x)}$$

5. Modus Tollens

If a statement implies another, and the second is false, then the first must also be false.

$$\frac{\begin{array}{l} p(x) \rightarrow q(x) \\ \neg q(x) \end{array}}{\therefore \neg p(x)}$$

6. Rule of Disjunctive Syllogism

If one of the two OR statements is false, then the other must be true.

$$\frac{\begin{array}{l} p(x) \vee q(x) \\ \neg p(x) \end{array}}{\therefore q(x)}$$

7. Rule of Universal Specification

If a statement is true for all elements, then it is true for a particular element.

$$\frac{\forall x \, p(x) \quad a \in S}{\therefore p(a)}$$

Arguments and Validity

Problems

1. Test the validity of the argument

Statement:

All cricketers are rich.

Sachin is a cricketer.

$\therefore$ Sachin is rich.

Solution:

- S = set of all cricketers
- a = Sachin
- $p(x)$: x is rich

Mathematical Form:

$$\frac{\forall x \in S, \, p(x) \quad a \in S}{\therefore p(a)}$$

Justification:

By the **Rule of Universal Specification**, if a statement is true for all elements of a set, then it is true for any particular element. **Conclusion:**

Hence, Sachin is rich and the argument is valid.

2. Test the validity of the argument Statement:

No engineering students are bad in studies.

Abhi is not bad in studies.

$\therefore$ Abhi is an engineering student

Solution:

- S = set of all engineering students
- a = Abhi
- $p(x)$: x is an engineering student
- $q(x)$: x is bad in studies

Mathematical Form:

$$\begin{array}{l} \forall x, [p(x) \rightarrow \neg q(x)] \\ \neg q(a) \\ \hline \therefore p(a) ? \end{array}$$

Explanation:

From the given statements,

$$p(x) \rightarrow \neg q(x)$$

means: If a person is an engineering student, then they are not bad in studies.

But we are given:

$$\neg q(a)$$

(Abhi is not bad in studies)

This does **not** guarantee that Abhi is an engineering student.

Reason:

This is a case of **fallacy of converse** (invalid reasoning):

$$p(x) \rightarrow \neg q(x) \not\Rightarrow \neg q(x) \rightarrow p(x)$$

Conclusion:

The given argument is invalid.

3. Test the validity of the argument Statement:

If a triangle has two equal sides, then it is isosceles.

If a triangle is isosceles, then it has two equal angles.

Triangle ABC does not have two equal angles.

$\therefore$ Triangle ABC does not have two equal sides

Solution:**Let:**

- x = triangle (ABC)
- $p(x)$: x has two equal sides
- $q(x)$: x is isosceles
- $r(x)$: x has two equal angles

Mathematical Form:

$$p(x) \rightarrow q(x)$$

$$q(x) \rightarrow r(x)$$

$$\neg r(a)$$

$$\therefore \neg p(a)$$

Proof:

From

$$p(x) \rightarrow q(x), \quad q(x) \rightarrow r(x)$$

$$\therefore p(x) \rightarrow r(x) \quad (\text{Syllogism})$$

Now,

$$p(a) \rightarrow r(a)$$

$$\neg r(a)$$

$$\therefore \neg p(a) \quad (\text{Modus Tollens})$$

Justification:

Using **Rule of Syllogism** and **Modus Tollens**, we derive the conclusion logically.

Conclusion:

Hence, triangle ABC does not have two equal sides and the argument is valid.

4. Test the validity of the argument Statement:

No engineering students of I sem study logic.

Abhi is an engineering student who studies logic.

$\therefore$ Abhi is not in I sem

Solution:

- S = set of all engineering students
- a = Abhi
- $p(x)$: x is an engineering student of I sem
- $q(x)$: x is an engineering student
- $r(x)$: x studies logic

Mathematical Form:

$$\begin{array}{l} \forall x, [(p(x) \wedge q(x)) \rightarrow \neg r(x)] \\ q(a) \wedge r(a) \\ \hline \therefore \neg p(a) \end{array}$$

Proof:

$$\begin{array}{l} [(p(a) \wedge q(a)) \rightarrow \neg r(a)] \wedge r(a) \\ \therefore \neg(p(a) \wedge q(a)) \quad (\text{Modus Tollens}) \\ \therefore \neg p(a) \vee \neg q(a) \quad (\text{De Morgan's Law}) \\ \therefore \neg q(a) \vee \neg p(a) \quad (\text{Commutative Law}) \\ \therefore \neg p(a) \quad (\text{Conjunctive Simplification}) \end{array}$$

Justification:

Using **Modus Tollens**, **De Morgan's Law**, and **Simplification**, we derive the result.

Conclusion:

Hence, Abhi is not in I sem and the argument is valid.

Truth Values of Quantified Statements

Question:

Determine the truth value of each of the following quantified statements.

The universe of discourse is the set of all non-zero integers.

1. $\exists x, \exists y [xy = 1]$
2. $\exists x, \forall y [xy = 1]$
3. $\forall x, \exists y [xy = 1]$
4. $\exists x, \exists y [(2x + y = 5) \wedge (x - 3y = -8)]$
5. $\exists x, \exists y [(3x - y = 17) \wedge (2x + 4y = 3)]$

Answers:

1. **True**

Take $x = 1, y = 1$

$$xy = 1 \times 1 = 1$$

Hence, the statement is true.

2. **False**

For fixed $x = 1$, take $y = 2$

$$xy = 1 \times 2 = 2 \neq 1$$

Hence, the statement is false.

3. **False**

Take $x = 2$

$$2y = 1 \Rightarrow y = \frac{1}{2} \notin \mathbb{Z}$$

Hence, the statement is false.

4. **True**

$$2x + y = 5, \quad x - 3y = -8$$

Substitute $x = 1, y = 3$

$$2(1) + 3 = 5, \quad 1 - 3(3) = -8$$

Hence, the statement is true.

5. **False**

$$3x - y = 17, \quad 2x + 4y = 3$$

$$y = 3x - 17$$

$$2x + 4(3x - 17) = 3 \Rightarrow 14x = 71$$

$$x = \frac{71}{14} \notin \mathbb{Z}$$

Hence, the statement is false.

Question:

Determine the truth value of each of the following quantified statements.

The universe of discourse is the set of all non-zero integers.

1. $\exists x, \exists y [xy = 1]$
2. $\exists x, \forall y [xy = 1]$
3. $\forall x, \exists y [xy = 1]$
4. $\forall x, \exists y [xy = 1]$
5. $\exists x, \exists y [(2x + y = 5) \wedge (x - 3y = -8)]$
6. $\exists x, \exists y [(3x - y = 17) \wedge (2x + 4y = 3)]$

Answers:**1. True**

Take $x = 1, y = 1$

$$xy = 1 \times 1 = 1$$

Hence, the statement is true.

2. False

For fixed $x = 1$, take $y = 2$

$$xy = 1 \times 2 = 2 \neq 1$$

Hence, the statement is false.

3. False

Take $x = 2$

$$2y = 1 \Rightarrow y = \frac{1}{2} \notin \mathbb{Z}$$

Hence, the statement is false.

4. False

Same reasoning as above: for $x = 2$, no integer y satisfies $xy = 1$.

$$2y = 1 \Rightarrow y \notin \mathbb{Z}$$

Hence, the statement is false.

5. True

$$2x + y = 5, \quad x - 3y = -8$$

Take $x = 1, y = 3$

$$2(1) + 3 = 5, \quad 1 - 3(3) = -8$$

Hence, the statement is true.

6. False

$$\begin{aligned}
 3x - y &= 17, & 2x + 4y &= 3 \\
 y &= 3x - 17 \\
 2x + 4(3x - 17) &= 3 \Rightarrow 14x = 71 \\
 x &= \frac{71}{14} \notin \mathbb{Z}
 \end{aligned}$$

Hence, the statement is false.

Question:

For the universe of all integers, let $p(x)$, $q(x)$, $r(x)$, $s(x)$ and $t(x)$ denote:

- $p(x) : x > 0$
- $q(x) : x$ is even
- $r(x) : x$ is a perfect square
- $s(x) : x$ is divisible by 3
- $t(x) : x$ is divisible by 7

Express the following statements in symbolic form.

1. At least one integer is even.
2. There exists a positive integer that is even.
3. If x is even then x is not divisible by 3.
4. No even integer is divisible by 7.
5. There exists an even integer divisible by 3.
6. Every integer is either even or odd.

Answers:

1. $\exists x q(x)$
There exists at least one integer which is even.
2. $\exists x [p(x) \wedge q(x)]$
There exists a positive integer which is even.
3. $\forall x [q(x) \rightarrow \neg s(x)]$
If an integer is even, then it is not divisible by 3.
4. $\forall x [q(x) \rightarrow \neg t(x)]$
No even integer is divisible by 7.
5. $\exists x [q(x) \wedge s(x)]$
There exists an even integer divisible by 3.
6. $\forall x [q(x) \vee \neg q(x)]$
Every integer is either even or not even (odd).

Methods of Proof

Direct Proof

The direct method is used to prove a conditional statement of the form:

$$p \rightarrow q$$

Steps:

1. **Hypothesis:** Assume that p is true.
2. **Analysis:** Use definitions, algebraic manipulation, and known results.
3. **Conclusion:** Show that q is true.

Example: Prove that if n is an odd integer, then n^2 is odd.

Proof: Assume n is odd. Then $n = 2k + 1$ for some integer k .

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

Thus, n^2 is of the form $2m + 1$, so it is odd.

Conclusion: Hence, the statement is proved by direct proof.

Indirect Proof

An indirect proof is used when direct proof is difficult. It is based on proving the contrapositive.

Contrapositive:

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Steps:

1. Assume $\neg q$ is true.
2. Show that $\neg p$ is true.
3. Hence conclude $p \rightarrow q$ is true.

Example: Prove that if n^2 is odd, then n is odd.

Proof: We prove the contrapositive: If n is not odd (i.e., even), then n^2 is not odd.

Assume $n = 2k$ for some integer k .

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

Thus, n^2 is even (not odd).

Conclusion: Therefore, the contrapositive is true, hence the given statement is true.

Note:

- Direct proof starts from assumption and reaches conclusion.
- Indirect proof uses contrapositive when direct proof is hard.

Problems**1. Prove by (i) Direct Proof (ii) Indirect Proof (iii) Proof by Contradiction:**

If n is an odd integer, then $n + 9$ is an even integer.

Solution:**(i) Direct Proof:**

Assume that n is an odd integer. Then $n = 2k + 1$ for some integer k .

$$n + 9 = (2k + 1) + 9 = 2k + 10 = 2(k + 5)$$

Thus, $n + 9$ is of the form $2m$, so it is even.

Conclusion: Hence, the statement is proved by direct proof.

(ii) Indirect Proof:

We prove the contrapositive.

If $n + 9$ is not even, then n is not odd.

Assume $n + 9$ is not even (i.e., odd). Then $n + 9 = 2k + 1$ for some integer k .

$$n = 2k + 1 - 9 = 2k - 8 = 2(k - 4)$$

Thus, n is even (not odd).

Conclusion: Hence, the given statement is true by indirect proof.

(iii) Proof by Contradiction:

Assume that the given statement is false. That is, assume n is odd but $n + 9$ is not even.
Let $n = 2k + 1$.

$$n + 9 = 2k + 1 + 9 = 2k + 10 = 2(k + 5)$$

Thus, $n + 9$ is even, which contradicts the assumption.

Conclusion: Hence, the statement is true by contradiction.

2. For any two odd integers m and n , show that:

(i) $m + n$ is even

(ii) mn is odd

Solution:

Let m and n be two odd integers.

$$m = 2a + 1, \quad n = 2b + 1 \quad \text{where } a, b \in \mathbb{Z}$$

(i) $m + n$ is even

$$\begin{aligned} m + n &= (2a + 1) + (2b + 1) \\ &= 2a + 2b + 2 \\ &= 2(a + b + 1) \end{aligned}$$

Hence, $m + n$ is of the form $2k$, so it is even.

(ii) mn is odd

$$\begin{aligned} mn &= (2a + 1)(2b + 1) \\ &= 4ab + 2a + 2b + 1 \\ &= 2(2ab + a + b) + 1 \end{aligned}$$

Hence, mn is of the form $2k + 1$, so it is odd.

3. Prove that: If n^2 is odd, then n is odd (by indirect proof).

Solution:

We prove the contrapositive:

If n is not odd, then n^2 is not odd

Assume n is not odd $\Rightarrow n$ is even.

$$n = 2k, \quad k \in \mathbb{Z}$$

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

Thus, n^2 is even (not odd).

Conclusion: Hence, by contrapositive, if n^2 is odd then n is odd.

4. Give direct proofs of the statements:

(i) If k and l are odd, then $k + l$ is even

(ii) If k and l are odd, then kl is odd

Solution:

Let k and l be odd integers.

$$k = 2a + 1, \quad l = 2b + 1 \quad (a, b \in \mathbb{Z})$$

(i) $k + l$ is even

$$\begin{aligned} k + l &= (2a + 1) + (2b + 1) \\ &= 2a + 2b + 2 \\ &= 2(a + b + 1) \end{aligned}$$

Hence, $k + l$ is even.

(ii) kl is odd

$$\begin{aligned} kl &= (2a + 1)(2b + 1) \\ &= 4ab + 2a + 2b + 1 \\ &= 2(2ab + a + b) + 1 \end{aligned}$$

Hence, kl is odd.